MILITARY SPECIFICATION

CARTRIDGE, 25.4MM, DECOY: M839

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1. This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of chaff payload designated as Cartridge, 25.4MM, Decoy: M839, hereafter referred to as the decoy cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-C-63106	-	Countermeasure, Chaff: M-1
MIL-C-63108	-	Cartridge, Impulse: M796

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection by Attributes

PSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extend specified herein.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

PRODUCT AND PACKAGING DRAWINGS

9311402	-	Countermeasure, Chaff: M1
9311660	-	Cartridge, Impulse: M796
9329609	-	Cartridge, 25.4MM, Decoy: M839
9329775	-	Packing and Marking for Container: PA19 for Cartridge, 25.4MM, Decoy: M839
9329776	-	Packing and Marking for Box, Exterior for Cartridge, 25.4MM, Decoy: M839

INSPECTION EQUIPMENT DRAWINGS

9311451	-	Payload Module Assembly
9311434	-	Dispenser Assembly
9311428	-	Breech Plate Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (Dwg.) 9329609 and with all requirements in applicable specifications.

3.2.1 Impulse cartridge. The assembly shall be inspected and tested in accordance with MIL-C-63108.

3.2.2 Countermeasure, Chaff: M1. The assembly shall be inspected and tested in accordance with MIL-C-63106.

Source: https://assist.dla.mil -- Downloaded: 20260-23T02:41Z
Check the source to verify that this is the current version before use.

3.3 Transportation vibration. The decoy cartridge shall remain functional after being subjected to MIL-STD-810, Method 514.2, Procedure X, using curve AW.

3.4 Static firing test. The decoy cartridge shall be fired in accordance with 4.5.2. After firing, not more than 5% by weight of the chaff is permitted to remain in the cartridge.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. All parts shall be assembled and finished in a thorough, workmanlike manner and all assembly operations shall be correctly performed. They shall be free of chips, sharp edges, cracks, warpage, cold flow, chipped edges, dirt, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer. The first article sample shall consist of one hundred units. (See Table I).

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078 and 4.5 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63338A(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Cartridge, 25.4MM, Decoy: M839		1 OF 1		See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge, 25.4MM, Decoy: M839 (Dwg. 9329609)				
	Examination for defects	100		3.2	4.4.2.1
	Transportation vibration	(a)		3.3	4.5.1
	Static firing test	(a)		3.4	4.5.2

NOTES:

(a) To be performed on the above 100 cartridges in the sequence indicated.

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cartridge, 25.4MM, Decoy: M839				9329609
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Impulse cartridge missing		0.40%	3.2	Visual
102	Cartridge case damaged or end cap loose		0.40%	3.2	Visual
Minor					
201	Marking improper or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63338A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Packing and Marking for Container: PA19 for Cartridge, 25.4MM, Decoy: M839	1 OF 1	9329775
			NEXT HIGHER ASSEMBLY: 9329776

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of cartridges		0.40%	3.2	Visual
102	Any packing component missing (loose pack)		0.40%	3.2	Visual
103	Can dented or damaged		0.40%	3.2	Visual
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63338A(AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.3	Packing and Marking for Box, Exterior for Cartridge, 25.4MM, Decoy: M839		1 OF 1	9329776
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any filler missing (loose pack)		0.40%	3.2	Visual
102	Number of PA19 containers incorrect		0.40%	3.2	Visual
103	Strapping and sealing missing or incorrectly applied		0.40%	3.2	Visual
Minor					
201	Exterior markings missing, incorrect or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the First Article table.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Transportation vibration. One hundred decoy cartridges packed 50/metal Can and two Cans per wirebound wooden box shall be subjected to the environment as defined in paragraph 3.3. Following this test the Cartridges shall be examined for defects and testing in accordance with 4.5.2.

4.5.2 Static firing test. (See 3.4) After transportation vibration the decoy cartridges shall be fired from an approved fixture (Dispenser Assembly, Dwg. 9311434, Payload Module Assembly, Dwg. 9311451). Any chaff payload failing to eject and disperse after the cartridge has fired shall be cause for rejection of the First Article Sample. If five or less impulse cartridges fail to function, they may be replaced. If more than five duds occur, the sample shall be rejected.

5. PACKAGING

5.1 Packaging, Level A. Packaging shall be as specified on Dwg. 9329775.

5.2 Packing, Level A. Packing shall be as specified on Dwg. 9329776.

5.3 Marking, Level A. Marking shall be as specified on Dwg. 9329775 and 9329776.

5.4 Level B and C. Packaging, packing and marking for Level B and C shall be the same as Level A.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the Cartridge, 25.4MM, Decoy: M839.

6.2 Ordering data. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required, to Commander: US Army Armament Research and Development Center, ATTN: DRSMC-QAT-I (D), Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander: US Army Armament Research and Development Center, ATTN: DRSMC-QAT-M (D), Dover, New Jersey 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originaly prepared by these activities are now under the cognizance of ARDC.

6.6 Changes from Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-A940)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-63338A	CARTRIDGE, 25 .4MM, DECOY: M839

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:41Z
Check the source to verify that this is the current version before use.